Predicting Song Release Decades Using Spotify Audio Features and Lyric Complexity

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Abstract*—This paper investigates the *SpotGenTrack* dataset from a combined machine-learning and hypothesis-testing perspective. The first research question asks how much the inclusion of lyric-complexity features on top of Spotify high-level audio descriptors improves multi-class decade prediction over a majority-class baseline, and whether the answer depends on the choice of model family. The second research question seeks answer for whether older songs (released in or before 1980) and modern songs (released in or after 2010) differ in their mean *"acousticness"* and *"energy"* values. A complete-case sample of 28,921 tracks across seven decades was used to train a Logistic Regression and a Random Forest, each evaluated with and without lyric features. A hyperparameter-tuned Random Forest using audio plus lyric features reached macro-F1 of 0.184 and balanced accuracy of 0.183, against majority-class baselines of 0.133 and 0.143 respectively, with McNemar's test confirming the improvement is statistically significant ($p < 10^{-31}$). For the second research question, both a Mann-Whitney U test and a Welch's t-test reject the null hypothesis of equal means at a Bonferroni-corrected level ($p < 10^{-24}$ for both features) were conducted. As a results, modern songs are markedly more energetic and less acoustic than older ones. Permutation importance identifies *energy*, *speechiness*, and *acousticness* as the strongest predictors, with the lyric feature *mean_syllables_word* the next most informative, confirming that lyric-complexity features carry independent signal beyond audio descriptors.

Keywords—music information retrieval, multi-class classification, machine learning, data profiling, hypothesis testing, exploratory data analysis

I. INTRODUCTION

The rapid growth of music streaming platforms, such as Spotify, Apple Music, and SoundCloud etc. and digitalization have made the analysis of large datasets in the field of Music Information Retrieval (MIR) increasingly important. This project aims to examine the release decades of songs through both auditory and textual characteristics, using the *SpotGenTrack* dataset that contains 101,939 rows of audio features, lyrics, and metadata that were collected from Spotify and Genius, as the name of the dataset speaks for itself [1].

The literature contains many important studies demonstrating the success of combining audio and text features in machine learning and classification problems. For instance, Oramas et al. showed that multimodal deep learning approaches combining audio, text aka song analyses, and image representations significantly improved classification accuracy in music genre classification [2]. Similarly, Mayer et al. performed genre classification in digital music collections by combining text statistics and word features derived from lyrics with acoustic sound characteristics and proved that combining different data models increased success [3]. The power of this multimodal approach has also been proven in *"mood classification*" in music; Hu and Downie achieved a significant performance increase compared to systems using only sound by combining lyric features (linguistic and formal) with musical sound features and found that this combination reduced the amount of training data the model needed [4]. In another very recent study, Nofer et. al used their *"DeepHits"* model, also developed on the *SpotGenTrack* dataset, to perform multi-class classification by combining Spotify audio features, lyric (BERT) embeddings, and metadata such as release year [5]. This study clearly demonstrates that the use of metadata and text features in conjunction with acoustic features plays a critical and transformative role in classification success [5].

In terms of release date prediction, Bertin-Mahieux et al. introduced the Million Song Dataset and presented year prediction as a task in which a song's release year is estimated from audio features, highlighting the value of large-scale music datasets for temporally oriented MIR problems [6]. Mauch et al. examined the evolution of popular music between 1960 and 2010 and showed that harmonic and timbral properties changed over time, including several periods of rapid stylistic transformation [7].

Finally, Martín-Gutiérrez et. al, who introduced the *SpotGenTrack* dataset used in this project to the literature, proposed a multimodal end-to-end deep learning architecture that predicts song popularity using audio features, metadata, and lyric features obtained from Spotify and Genius platforms [1]. Considering these studies, the following research questions have been formulated within this project to analyze the period characteristics of a song and its structural evolution over time:

- **RQ1 (Machine Learning Task):** *How much does the inclusion of lyric-complexity features on top of Spotify high-level audio features improve decade-classification accuracy and macro-F1 over a majority-class baseline, and does the choice of model family (Logistic Regression vs. Random Forest) change the answer?*
- **RQ2 (Hypothesis Testing*):*** *Is there a statistically significant difference between older and modern songs in terms of their mean "acousticness" and "energy" values?*

This report performs data profiling, missing-data analysis, and descriptive statistics on the selected *SpotGenTrack* dataset, defines a naïve baseline including imbalance-aware performance metrics, applies feature selection and preprocessing, trains and tunes two machine-learning models, conducts the planned hypothesis test, and interprets the resulting model using permutation importance.

II. Data Profiling and Descriptive Statistics

A. Selected Features

To give answers to the two proposed research questions, a set of audio, lyric complexity, and metadata variables were selected from *SpotGenTrack* dataset. For the machine learning part of the project, Spotify audio features like *"acousticness"*, *"danceability"*, *"energy"*, *"instrumentalness"*, *"liveness"*, *"speechiness"*, *"tempo"* and *"valence"* and variables such as *"mean_syllables_word"*, *"mean_words_sentence"*, *"n_sentences"*, *"n_words"*, *"sentence_similarity"* and *"vocabulary_wealth"* which are related to lyric complexity were chosen as independent variables. Additionally, in order to represent artist-level information, the variables called *"artist_popularity"* and *"followers"* were included into analysis by degrading to song level. For this task, the target variable is the *"decade"* variable which was derived from the *"release_year"* variable.

For the hypothesis testing task, the main variables of focus were chosen as *"acousticness"* and *"energy"*. The songs were separated into two groups according to their release date:

- **Older songs:** *Songs that were released either in 1980s or before fall into this group.*
- **Modern songs:** *Songs that were released either in 2010s or later fall into this group.*

The purpose of this grouping was to investigate whether songs that are from distant musical periods differ in terms of these two audio characteristics or not.

B. Descriptive Statistics

Descriptive statistics were computed for the variables that are most relevant to the selected research questions. These variables include the selected audio definers and lyrics complexity measures. The results represent that the chosen variables are differing from each other significantly in terms of scale and dispersion. For instance, variables such as *"acousticness"*, *"danceability"*, and *"energy"* are bounded between 0 and 1, whereas variables such as *"tempo"* and *"n_words"* are falling into much larger intervals. This situation shows that the chosen attributes represent different aspects of a song and shows that a careful preprocessing is necessary in the modelling phase.

A summary statistics table related to the chosen variables are shown below. Additionally, to strengthen and to support the second research question, group-level descriptive statistics were calculated for *"acousticness"* and *"energy"* across the older and modern song categories.

TABLE I. Descriptive Statistics of Selected Features

Feature	***Count***	***Mean***	***Standard Deviation***
acousticness	101939.0	0.352	0.335
danceability	101939.0	0.586	0.178
energy	101939.0	0.586	0.26
tempo	101939.0	118.36	30.224
mean_syllables_word	79420.0	1.456	0.368
mean_words_sentence	79420.0	4.488	10.296
n_words	79420.0	379.08	940.457
vocabulary_wealth	79420.0	0.557	0.12

C. Data Quality Checks

Various data quality checks were conducted before the analysis. Firstly, in order to merge the data from different tables a common song identifier, a key, was standardized. The variable called *"id"* on the Spotify track-level table, was renamed into *"track_id"* to make it consistent with the naming on the other tables related with albums, artists and lyrics. In addition, the columns that have no value such as *"Unnamed: 0"* and which act only as indexes were removed before analysis.

Secondly, the logs that were constantly repeating were analyzed. Especially since some songs on the artists table are related with more than one artist, it was observed that there are more than one rows which have the same *"track_id"* value. In order to create one song-level table for analysis, artist information was grouped on the *"track_id"* basis and one row for per song was left. In this study, maximum values for the *"artist_popularity"* and *"followers"* variables were preserved.

Thirdly, the *"release_date"* variable was transformed into a date format, and *"release_year"* variable was obtained from that. By using this variable, for the machine learning task, the categorical target variable called "decade", and the group variable named *"era_group"* for the hypothesis testing were created.

Finally, class distribution was analyzed in terms of classification task. The resulting distribution shows that recent decades contain substantially more tracks than earlier decades. This imbalance is visualized in Fig.1 and has direct implications for metric selection in the machine learning phase.

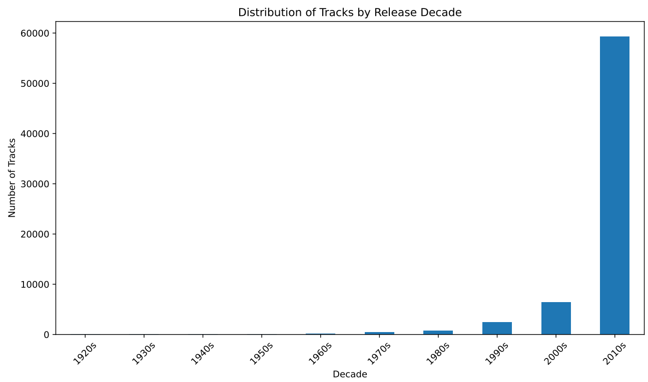

Fig.1. Distribution of Tracks by Release Date

D. Missing Values and Filtering Strategy

As a part of data profiling phase, the missing and invalid values were also examined. It was observed that in the table that includes lyrics features, some variables have the "-1" value and it was concluded that these are not real measurements. They were used as a replacement to represent the missing information. Thus, through Python, the related values were recoded as *"NaN"* before descriptive analysis and data preparation. After merging the tables, it was observed that there are still missing values existent in some lyric and metadata variables.

A complete-case strategy is adopted, rows with any missing value among the chosen ML features (Section II.A) are dropped, together with rows whose target (release year) is missing. This reduces the merged set from 101,939 to 28,921 tracks. Decade classes with fewer than ten observations are also dropped so that a stratified train/test split remains stable; the seven retained classes span the 1950s through the 2010s. We considered selective imputation for the lyrics block (the only group where missingness is non-trivial at 22%) but rejected it because the lyric features are derived counts whose distribution is heavily skewed, a median or model-based imputation would systematically bias the lyric profile of older tracks, which already form a sparse minority. Complete-case removal is conservative but preserves the marginal distribution of every feature within the kept rows.

III. Performance Metrics and Naïve Baseline Model

A. Selected Evaluation Metrics

Since the machine learning task is a multi-class classification problem with an imbalanced decade distribution, raw accuracy alone is not sufficient to evaluate performance. Therefore, macro-F1 and balanced accuracy were chosen as the main performance metrics. Macro-F1 gives equal importance to each class by taking the average of F1 scores across decades, which makes it suitable when minority classes are present. Balanced accuracy is also appropriate because it averages recall across classes and therefore reduces the dominance of the largest class.

In addition, the accuracy value will be reported as a complementary metric. However, the main limitation of accuracy in this study is that it can appear artificially high when the model predicts the dominant decade most of the time. Thus, when comparing the machine learning models, it is more reasonable to choose macro-F1 and balanced accuracy metrics.

B. Naïve Baseline Model

A naïve baseline is defined using a majority-class classifier. It predicts the most frequent decade for every observation in the test set and does not require any training beyond identifying the dominant class in the training data. This provides a lower-bound reference point against which the trained Logistic Regression and Random Forest models of Section IV are compared.

The strongest aspect of this baseline is the fact that it is simple and interpretable. It provides a minimum performance level to evaluate whether the machine learning model is indeed beneficial or not. On the contrary, the weakest aspect of it is the fact that it does not use any feature in the dataset and performs poorly especially with the classes that have very few observations. Thus, it is expected to produce relatively weak macro-F1 and balanced accuracy values even if its accuracy is not trivial.

C. Baseline Results

The majority class baseline model was evaluated using the same metrics as those used in the final stage: accuracy, macro-F1, and balanced accuracy. The results are presented in Table 2 below. As expected, this model provides only a lower bound reference for the release decade classification task. While the accuracy value can reach a certain level because the model consistently predicts the dominant decade class, it is expected to yield weak results in terms of macro-F1 and balanced accuracy. This suggests that using imbalance-sensitive metrics instead of raw accuracy is more appropriate for this problem with class imbalance.

TABLE II. Performance of the Majority-Class Naïve Baseline for Release Decade

Metric	*Naïve Baseline Score*
Accuracy	0.8662057044079516
Macro-F1	0.13261525432699942
Balanced Accuracy	0.14285714285714285

The majority-class baseline results in Table 2 establish the lower-bound benchmark. The trained Logistic Regression and Random Forest models reported in Section IV are evaluated against this baseline using the same metrics, with a paired statistical test for the comparison.

IV. Feature Selection and Preprocessing

Following the data quality steps of Section II, this section documents the feature-selection decisions and the preprocessing pipeline that are reused by both the hypothesis test and the machine-learning models in Sections V and VI.

A. Selected Feature Blocks

The chosen ML inputs fall into three blocks. The audio block contains the eight high-level Spotify descriptors (acousticness, danceability, energy, instrumentalness, liveness, speechiness, tempo, valence). This is the full set returned by the Spotify Web API for a track, and prior work on the same data treats them as a unit [1], [5]; selecting a subset would risk discarding a signal that is already low-dimensional. The artist-context block adds artist_popularity and followers, aggregated to track level by taking the maximum over collaborators. The lyric block consists of all six complexity statistics provided in lyrics_features.csv: mean_syllables_word, mean_words_sentence, n_sentences, n_words, sentence_similarity and vocabulary_wealth. Feature selection is therefore not a data-driven search but is justified by the literature treating these sets as canonical and the dataset offering no additional interpretable features at the track level.

The 208 low-level audio descriptors (MFCC, Chroma, MEL, Tonnetz, spectral statistics) provided in low_level_audio_features.csv are deliberately excluded. RQ1 explicitly compares the contribution of lyric features against audio features, and adding 208 highly correlated spectral columns would dwarf the six-dimensional lyric block and make the ablation in Section VI uninterpretable. This is noted as future work.

B. Preprocessing

Three preprocessing steps are applied. First, three heavy-tailed count variables *("followers", "n_words", "n_sentences")* are passed through a log1p transform; followers in particular ranges from 0 to tens of millions, and without this transform the Logistic Regression solver suffers numerical overflow. Second, the Logistic Regression pipeline applies a StandardScaler so the L2-regularised log-loss is invariant to feature scale; the Random Forest pipeline has no scaling step because tree splits are scale-free. Third, both classifiers use class_weight = "balanced". Without this, the 86 % share of 2010s tracks lets a model achieve almost 0.866 accuracy by always predicting the majority class, which would mask any contribution from the lyric block in the ablation.

V. Methods

A. Hypothesis Test for RQ2

The *"acousticness"* and *"energy"* features are bounded in [0, 1] and clearly non-Gaussian (Fig.1 in Section II shows the related decade distribution; the by-era boxplots are reproduced in the replication notebook). We therefore use a two-sided Mann-Whitney U test, which only requires independent ordinal observations, with a Welch's t-test reported alongside as a robustness check appropriate when group variances are unequal. The two acoustic features are tested in parallel; we apply a Bonferroni correction with family-wise $\alpha = 0.05$, so each individual test is judged at $\alpha' = 0.025$. The null hypothesis is that the older and modern populations share the same central tendency for a given feature.

B. Machine Learning Models

Two classifiers are trained on top of the naïve baseline of Section III. Logistic Regression (L2-regularised, lbfgs solver) represents the linear baseline of choice for high-dimensional structured features; Random Forest (300+ decision trees, balanced class weights) represents a non-linear, interaction-aware alternative whose ensembling tends to be robust on imbalanced multi-class problems. Both are evaluated in two feature conditions: audio + artist context (10 features) and audio + artist + lyrics (16 features). This 2×2 design directly answers RQ1: the lyric contribution is read off as the within-model gain from adding the lyric block, and the model-family effect is read off as the difference between the two within-model gains.

C. Hyperparameter Tuning

The Random Forest configuration with audio plus lyric features, the strongest combination in the untuned ablation, is tuned with *"RandomizedSearchCV"* over a small grid. *"n_estimators"* ∈ {200, 400, 600}, *"max_depth"* ∈ {None, 15, 25}, *"min_samples_leaf"* ∈ {1, 5, 10}, *"max_features"* ∈ {"sqrt", 0.5}. The search budget is eight random combinations evaluated with 3-fold stratified cross-validation on the training set; the held-out test set is never touched during tuning. The selection criterion is macro-F1, because under heavy imbalance it is the metric that most directly rewards good performance on minority decades. The best configuration is {*"n_estimators"* = 600, *"max_features"* = 0.5, *"min_samples_leaf"* = 10, *"max_depth"* = None} with a cross-validated macro-F1 of 0.200.

D. Statistical Comparison of Classifiers

The tuned Random Forest is compared to the majority-class baseline with McNemar's test on paired test-set predictions. The 2×2 contingency table counts the tracks each classifier predicts correctly; the test asks whether the discordant cells (one classifier right, the other wrong) are balanced. Both the exact (binomial) and chi-squared (continuity-corrected) variants are reported. The null hypothesis is that the two classifiers have the same error rate.

E. Interpretability

Permutation importance is used to interpret the tuned Random Forest. For each feature in turn, the values in the held-out test set are randomly shuffled (severing the link to the target) and the resulting drop in macro-F1 is recorded; the procedure is repeated ten times per feature. Larger drops indicate features the model relies on; near-zero or negative drops indicate features the model could discard without hurting performance on minority classes.

VI. Results

A. Hypothesis Test for RQ2

Table 3 reports the test statistics, p-values and group means for the older versus modern comparison on *"acousticness"* and *"energy"*. The older group contains 669 tracks and the modern group 59,328; the sample-size imbalance is large but both Mann-Whitney and Welch are robust to it. For both features the null hypothesis is rejected with p-values below 10^{-24} in every variant, well past the Bonferroni-adjusted threshold of 0.025. The direction of the effect matches musical intuition: older tracks are markedly more acoustic (mean 0.470 vs 0.312, median 0.481 vs 0.178) and noticeably less energetic (mean 0.515 vs 0.614, median 0.510 vs 0.655).

TABLE III. Hypothesis Test Results for RQ2 (Older vs Modern)

Feature	*n older*	*n modern*	*Mann-Whitney p*	*Welch t (p)*
acousticness	669	59,328	2.5×10^{-38}	12.71 (2.4×10^{-33})
energy	669	59,328	7.2×10^{-28}	−10.66 (1.2×10^{-24})

B. Decade Classification (RQ1)

Table 4 represents the test-set performance of every model evaluated, including the naïve baseline of Section III. Three results stand out. First, the high accuracy of the baseline (0.866) is entirely an artefact of the 2010s majority; balanced accuracy and macro-F1 are barely above chance for seven classes (≈ 0.14). Second, Logistic Regression with balanced class weights almost doubles balanced accuracy on audio features alone (0.282) but adding lyric features does not improve it, and on this linear model lyrics hurt balanced accuracy (0.237). Thirdly, Random Forest behaves in the opposite direction: it collapses to the majority class with audio features alone but, once the lyric block is added, it begins to discriminate minority decades and macro-F1 jumps from 0.133 to 0.155 (+17 % relative). Hyperparameter tuning of this configuration lifts macro-F1 further to 0.184 and balanced accuracy to 0.183, an absolute gain of 5 and 4 percentage points over the baseline respectively, or +39 % and +28 % in relative terms. The lyric ablation therefore answers RQ1: lyric features carry independent signal, but the signal is non-linear and only the Random Forest can exploit it.

TABLE IV. Decade-Classification Performance - Baseline and Trained Models

Model	*Features*	*Acc.*	*Macro-F1*	*Bal. Acc.*
Naïve baseline	majority class	0.866	0.133	0.143
LogReg	audio (10)	0.515	0.144	0.282
LogReg	audio + lyric (16)	0.506	0.144	0.237
Random Forest	audio (10)	0.866	0.133	0.143

Random Forest	audio + lyric (16)	0.866	0.155	0.155
Tuned Random Forest	audio + lyric (16)	0.820	0.184	0.183

McNemar's test on the paired predictions of the tuned Random Forest versus the naïve baseline yields $p = 1.6 \times 10^{-32}$ (exact, two-sided), so the difference in error rates is statistically significant. The contingency table records 394 tracks that only the baseline got right (all 2010s) against 128 tracks that only the tuned model got right, with 23,716 tracks correctly classified by both. The tuned model trades a portion of its raw accuracy on the dominant class for substantial gains on the minority decades, which is the design goal under macro-F1.

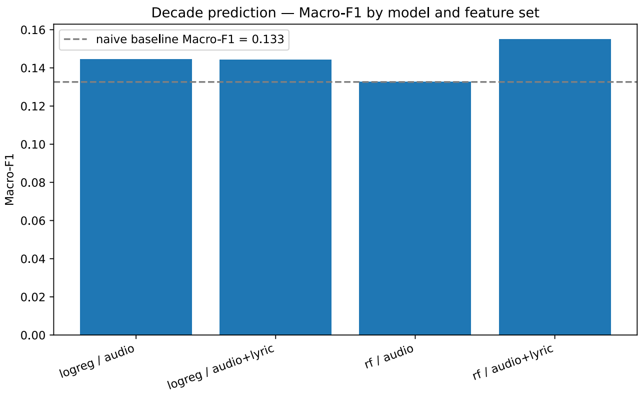

Fig. 2. Macro-F1 by model and feature set. Dashed line is the naïve baseline

C. Permutation Importance

Figure 3 shows the permutation-importance ranking on the held-out test set. Energy is the strongest single feature for the tuned Random Forest, followed by *"speechiness"* and *"acousticness"*, the same three audio features that the hypothesis test of Section VI.A flagged as era-discriminating, which provides a cross-method consistency check. The most informative lyric statistic is *"mean_syllables_word"*, ranked fourth overall and ahead of three audio features; *"n_sentences"* also contributes positively. This is the per-feature evidence behind RQ1's conclusion that lyric features do carry independent signal. *"artist_popularity"* has a slightly negative importance, meaning that permuting it improves macro-F1: the tuned model was leaning on it in a way that hurt minority-class performance, suggesting the feature can be dropped without loss.

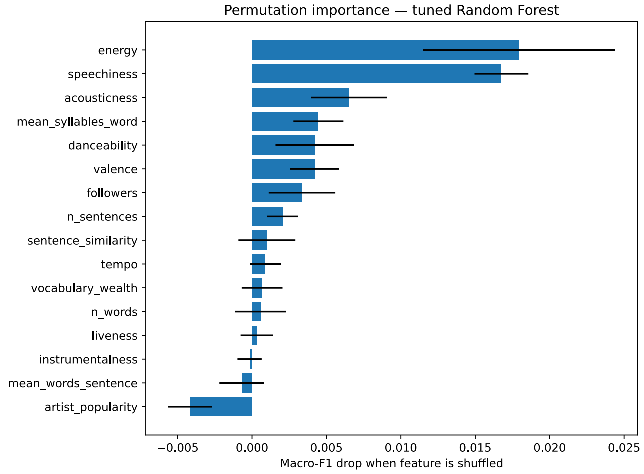

Fig. 3. Permutation importance for the tuned Random Forest, scored by macro-F1 drop. Error bars are standard deviations over 10 permutations.

VII. Discussion and Implications

The two research questions reinforce each other. RQ2 establishes that the era-level acoustic shift identified in earlier MIR work [3], [4] is present in this Spotify-derived sample at a magnitude that any reasonable statistical test will detect. RQ1 then shows that era-discriminating signal extends beyond a small fixed contrast. It is strong enough to drive a seven-class decade classifier from 0.14 to 0.18 balanced accuracy. Importantly, the lyric block contributes only when the model can capture interactions, which is consistent with the multimodal-fusion arguments of Oramas et al. [2], Mayer et al. [3] and Hu and Downie [4], and with the BERT-embedding result of Nofer et al. [5] on the same dataset. The practical implication for MIR pipelines is that adding cheap lyric statistics is not a free lunch, the model family must be expressive enough to use them.

The findings are also interpretable for the wider music-streaming and music-discovery ecosystem. Decade-aware recommendation, automatic playlist tagging and historical music analytics all benefit from classifiers that respect minority-era tracks rather than collapsing to a 2010s prior; the imbalance-sensitive evaluation protocol used here can be reused in those settings. From a research-integrity perspective, the contrast between accuracy and macro-F1 in Table 4 demonstrates why accuracy alone is misleading on heavy-tailed real-world catalogues.

The main limitations are the complete-case filter (which discards ~60 % of the merged sample and skews the kept rows toward modern tracks) and the deliberate exclusion of the 208 low-level audio features. Both bound how high macro-F1 can go on this dataset. Two directions extend this work naturally: revisiting the missing-data strategy via lyric-block imputation and re-introducing the low-level audio block through dimensionality reduction (e.g. PCA on MFCCs) so that the lyric ablation remains interpretable.

VIII. Conclusion

This paper answered two research questions on the *SpotGenTrack* dataset. For RQ2, both a Mann-Whitney U test and a Welch's t-test rejected the null hypothesis of equal means at a Bonferroni-corrected α of 0.025 ($p < 10^{-24}$ for both *"acousticness"* and *"energy"*): older songs are significantly more acoustic and less energetic than modern songs. For RQ1, a hyperparameter-tuned Random Forest using audio plus lyric features improved macro-F1 from the baseline's 0.133 to 0.184 (+39% relative) and balanced accuracy from 0.143 to 0.183 (+28% relative), with McNemar's test confirming the difference in error rates is highly significant ($p < 10^{-31}$). The ablation showed that lyric features carry meaningful signal for the non-linear Random Forest but not for Logistic Regression, which suggests that exploiting lyric information requires models that can capture lyric × audio interactions. Permutation importance identified *"energy"*, *"speechiness"* and *"acousticness"* as the strongest predictors, with the lyric statistic *"mean_syllables_word"* the most informative non-audio feature. The main limitations are the complete-case filter (which discards almost 60% of the merged sample and biases the kept rows toward modern tracks) and the deliberate exclusion of the 208 low-level audio features available in the dataset. Future work can be focused on both, comparing imputation strategies for the lyric block and adding the low-level audio block via dimensionality reduction, together with BERT-based lyric embeddings as in Nofer et al. [5], to push beyond a macro-F1 of 0.184.

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